

Steven M. Cutchin

Associate Professor of Computer Science

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Education

- **Ph.D., Computer Science**, Purdue University, August 2000.
Thesis: *Flexible User Interface Coupling with Operation Transformation*.
- **M.S., Computer Science**, Purdue University, May 1993.
- **B.S., Computer Science and Mathematics** (double major), Purdue University, May 1990.

Professional Experience

- **Associate Professor**, Department of Computer Science, Boise State University *Aug 2013–Present*
- **Faculty Director, Research Computing**, Boise State University *Aug 2016–May 2023*
- **Manager, Supercomputer & Visualization/VR Laboratories**, King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia *Nov 2008–June 2013*
- **Assistant Director**, San Diego Supercomputer Center (Visualization Research & Production) *June 2001–Nov 2008*
- **Visualization Scientist**, San Diego Supercomputer Center, UCSD *Nov 2000–June 2001*
- **Senior Software Engineer**, Walt Disney Feature Animation, Burbank, CA *Nov 1999–Nov 2000*
- **Research Scientist**, Texas Institute for Computational and Applied Mathematics, UT Austin *May 1997–Nov 1999*
- **Research Assistant**, Department of Computer Sciences, Purdue University *Aug 1991–May 1997*
- **Summer Intern**, Institute for Defense Analysis, Washington, D.C. *Summer 1995*
- **Software Engineer**, Intergraph Corporation, Huntsville, AL *June 1990–Aug 1991*

Research Interests

Immersive extended reality (XR) for education, perceptual training, and laboratory safety; visualization for blockchain / cryptocurrency security (anomaly, outlier, and wash-trading detection); topological and geometric methods for memory consolidation in AI / LLM agents; photo-realistic immersive capture and interaction; ultra-high-resolution display environments; terascale computational visualization; high dynamic range lighting; collaborative virtual environments; and serious games for science education.

Awards and Honors

- Project Grant, Idaho Commission on the Arts — XR drawing / perceptual training in arts education
2025
- IEEE VR Conference certificate of attendance *Mar 2022*
- Invited Member, Oculus NextGen Academic Research Program *Nov 2017*
- KAUST Seed Fund Award for product development: Photo-real 3D Immersion *Mar 2012*
- Best Visualization Display Award, TeraGrid Conference *2008*
- Certificate of Recognition for Outstanding Accomplishment in Assistant Teaching, Purdue ACM Student Chapter *1995*

Research Grants and Major Funding

- **PI**, *ChemVR: Extended Reality for Chemistry Laboratory Safety Training*, National Science Foundation, Award #2303007 (Boise State, USC, Chapman, LMU) *2023–Present*
- **PI / Faculty Lead**, *XR Perceptual Training in Arts Education*, Idaho Commission on the Arts (OSP #12209), \$10,000 *2025*
- **Faculty Mentor**, *Blockchain Research Experience for Undergraduates (BREU) Site*, NSF, Award #2349042 *2024–2025*
- **Co-Lead**, School of Arts Immersive Visualization Facility, Boise State University, \$2,000,000 *2019, 5 yr*
- **PI**, Borah Cyberinfrastructure Computational Facility, Boise State & Idaho Power, \$1,500,000 *2020, 6 yr*
- **co-PI**, CC*DNI Engineer: CI Engineer to Enhance Collaborative Research, NSF, \$381,221 *2015, 3 yr*
- **co-PI**, Precision Ag—Crop Yields Using IoT & Data Science, Idaho Dept. of Commerce, \$335,969 *2015, 1 yr*
- **co-PI**, iSEED (CyberInfrastructure Visualization for EPSCoR MILES), \$100,000 *2014, 1 yr*
- **PI**, VR Volume Visualization, Idaho National Laboratory / Battelle Energy Alliance, \$60,000 *2014, 1 yr*
- **co-PI**, KAUST-EPFL Brain Alliance Supercomputing Research Project, \$1M/yr *2012, 2 yr*
- **co-PI**, GSFC High Interactivity Visualization Software for Large Data, NASA SBIR Phase 2, \$132,000 *2007, 1 yr*
- **co-PI**, Collaborative Analysis Management System, NIH SBIR Phase I (Biotrue), \$132,000 *2006, 1 yr*
- **Project Manager**, IT-E3Tools (ITEST, Award #ESI-0624672) *2006*

- **Manager**, KAUST Supercomputing Facility, \$6M/yr *2010–2012*
- **Manager**, KAUST Visualization and Virtual Reality Facility, \$4M/yr *2008–2012*

Courses Taught

- CS 410/510 — Introduction to Databases, Boise State University *Fall 2026*
- CS 464/564 — Computer Graphics / Introduction to Data Visualization (grad cross-list) *Fall 2025*
- CS 564 — Visualization Techniques / AI-Driven Mixed Visualization Dashboards
- CS 402 — Introduction to Mobile Development (React Native) *Spring 2025*
- CS 321 — Data Structures *Spring 2025*
- CS 253 — Introduction to Software Development in C
- CS 361 — Introduction to Theory of Computation
- VIP-VR — Immersive Virtual Reality (Vertically Integrated Projects; faculty lead)
- Maya I: Introduction to 3-D Modeling (ART-40298), UCSD; Scientific Visualization short courses, San Diego Supercomputer Center

Graduate Students

- **Rezvan Joshaghani**, Ph.D., Computer Science (committee chair; XR/VR in undergraduate science education) — current
- **Aayushi Rajput**, Ph.D., Computer Science (XR perceptual training in arts education; Meta Quest 3 passthrough) — current
- **Bhargavi Rao Bondada**, M.S., Computer Science — current
- Iker Vazquez Lopez; Mike Ramshaw; Yuan Li; Suadad Jasim (Dec 2014); Ujjwal Acharya — graduated

Undergraduate Research Students

- Adam Crayton, Archie Rauenhorst — Blockchain REU (Bitcoin transaction visualization) *2025*
- Ezekiel Pinkerton, Sophia Oyakhire — blockchain visualization *2024*
- Kenneth Miller, Afton Carlson, Jonathon Stohler (2016), Anh Trinh (2015), Jay Capcha, Michael Pleasance, Lois Baek, Huma Aatifi, Alex Decastro (Fall 2016), Ryan Bailey — graduated

Service and Committees

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- Member, Computing Ph.D. Admissions Committee *Fall 2025, Fall 2026*
- Member, Faculty Senate Grievance Committee, Boise State University *2025–2026*
- Member, Computer Science Department Outreach Committee *2026–Present*
- Member, Periodic Post-Tenure Review committees (faculty reviews) *2025*
- Co-Organizer / Chair, IEEE VR KELVAR Workshop (K-12+ Embodied Learning through VR/AR)
IEEE VR 2023, 2025, 2026; accepted 2027
- Member, Luminary (World Museum) Immersive Theatre Facility Committee *2016–Present*
- Steering Committee, Idaho Virtual Reality Technology Council *2016–Present*
- Steering Committee & All-Hands co-Chair, NSF West Big Data Hub *2016–Present*
- Director, Research Computing Division, Boise State Office of Information Technology *2016–2023*
- Chair, Dean of Engineering Evaluation Committee *Spring 2016*

Program Committees and Reviewing

- Program Committee, GRAPP (Computer Graphics Theory and Applications) *2025*
- Program Committee / Reviewer, GRIVAPP *2026, 2027*
- Reviewer, IEEE VR K-12+ Embodied Learning / KELVAR Workshops *2025, 2026*
- NSF Review Panel, Directorate for Computer and Information Science and Engineering (CISE) *2015, 2020*
- NSF Office of Cyberinfrastructure, Strategic Technologies for Cyberinfrastructure *2007*
- NIH/NSF Workshop on Visualization Research Challenges; Modeling, Simulation & Gaming Technologies Applied to Education *2004*

Selected Presentations

- NSF Grantees Poster Session, ASEE Annual Conference (NSF #2303007) *2025*
- ChemVR project outcomes, ASEE Annual Conference & Exposition *2025*
- VR Chemistry Trainer / Chemical Laboratory Safety, COEN Undergraduate Lightning Talks, Boise State *2025*
- Blockchain Visualization, Blockchain REU Research Symposium, Boise State *2025*

- Immersive VR Experiences at Boise State, CAAV Conference for Visualization & VR Professionals, Villanova University *Nov 2018*
- Teaching CS Students About Algorithm Bias — ACRL (2020); invited talks, Cal State LA (2020) & USC (2019)
- The World Museum at Boise State, 12th International Conference on the Inclusive Museum, Buenos Aires *Nov 2019*
- Using XR to Teach Chemical Lab Safety, ACS Fall National Meeting, San Diego *Aug 2019*

Patents

- System and Method for Generating Three-Dimensional Panoramic Images. Publication No. US20130057643.

Selected Works in Media

- *Digital Recreation of a Seven-Story Building Shake During an Earthquake* — IEEE Vis Scientific Animation Theatre, Sacramento *Nov 2007*
- *Recreation of the 1906 San Francisco Earthquake* — 100th Anniversary Earthquake Conference, San Francisco *Apr 2006*
- *LA's Future Quake* (TeraShake & Puente Hills visualizations) — National Geographic Channel *Sep 2006*
- *Evolution of the Universe* — SIGGRAPH DomeFest, Los Angeles *Aug 2005*
- Animations for *100 Greatest Discoveries in Chemistry* — Discovery Science Channel *2004*

Professional Memberships

Association for Computing Machinery (ACM); IEEE.

Publications

- [1] Aayushi Rajput and Steven Cutchin. Advancing perceptual training in arts education through immersive extended reality systems. In *IEEE VR KELVAR Workshop: K-12+ Embodied Learning through Virtual and Augmented Reality, IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, Daegu, Korea, 2026. Listed on KELVAR 2026 workshop program (Mar 2026).
- [2] Rezvan Joshaghani, Alessandra Winters, Yanyan Zong, and Steven Cutchin. Xr module for enhancing glove hygiene skills in first-year chemistry laboratories. *IEEE Transactions on Visualization and Computer Graphics*, 32(5):41654173, May 2026. Accepted and published (submission #1781). Confirmed by author.

- [3] Yanyan Zong, Shalini Ramachandran, Steven Cutchin, et al. The effect of virtual reality on chemistry laboratory anxiety and self-efficacy. In *2026 AERA Annual Meeting*. American Educational Research Association, 2026. Accepted Nov 2025; Cutchin co-author (confirmed). Confirm author order/full list.
- [4] Steven M. Cutchin. Iuse: Using extended reality (xr) to close equity gaps in laboratory training. 2026 ASEE Annual Conference & Exposition, 2026. Abstract accepted Nov 2025; full paper pending.
- [5] Cesar Osvaldo Alcantar, Gaby Dagher, and Steven Cutchin. Detecting wash trading in cryptocurrency. 2026.
- [6] Rezvan Joshaghani, Yanyan Zong, and Steven Cutchin. Virtual reality and the affective gap: Evidence for immersive technology as transitional support in undergraduate science education. In *2026 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 248–253. IEEE, 2026.
- [7] Steven Cutchin and Aayushi Rajput. Advancing perceptual training in arts education through immersive virtual reality systems. 31st ACM Symposium on Virtual Reality Software and Technology (VRST '25), Montréal, Canada, 2025. Poster/Demo, accepted and presented. Confirmed by author.
- [8] Adam Crayton, Gaby Dagher, Archie Rauenhorst, and Steven Cutchin. Bitcoin visualization of outliers in behavior and transactions using band depth analysis. 2025.
- [9] Ezekiel Pinkerton, Sophia Oyakhire, Gaby Dagher, and Steve Cutchin. Illuminating the blockchain: Visualizing bitcoin transactions, anomalies, and dark web connections. Idaho Conference on Undergraduate Research (ICUR), Boise, ID, 2024. Poster/presentation; NSF award #2349042. Indexed at Boise State Experts / ScholarWorks.
- [10] Steven Cutchin, Erica Southgate, Jerry Alan Fails, and Manoela Milena Oliveira da Silva. Workshop: Ieee vr kelvar workshop: K-12+ embodied learning through virtual and augmented reality (8th annual workshop). In *2023 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, page 507. IEEE, 2023. Workshop organizer contribution; verified via DBLP.
- [11] HR Spero, I Vazquez-Lopez, K Miller, R Joshaghani, S Cutchin, and J Enterkine. Drones, virtual reality, and modeling: communicating catastrophic dam failure. *International Journal of Digital Earth*, 15(1):585–605, 2022.
- [12] Roberto Luna-Garcia, Gaby Dagher, Chabeli Castaño-Arango, and Steven Cutchin. Blockchain visualization. 2022.
- [13] Steven Cutchin. Mining online art museums for research and education. In *Faculty Research Lightning Talks, Boise State University*, 2022.
- [14] Chábeli Castaño Arango, Roberto Luna-Garcia, Steven Cutchin, and Gaby G Dagher. Towards deanonymization of mixing services in bitcoin. In *2022 IEEE 1st Global Emerging Technology Blockchain Forum: Blockchain & Beyond (iGETblockchain)*, pages 1–6. IEEE, 2022.
- [15] Dylan Eck, Adam Torek, Steven Cutchin, and Gaby G. Dagher. Diffusion: Analysis of many-to-many transactions in bitcoin. In *2021 IEEE International Conference on Blockchain (Blockchain)*, pages 388–393. IEEE, 2021. Verified via DBLP.
- [16] Shalini Ramachandran, Steven Matthew Cutchin, and Sheree Fu. Raising algorithm bias awareness among computer science students through library and computer science instruction. In

2021 ASEE Virtual Annual Conference Content Access. American Society for Engineering Education, 2021. Distinct from cutchinfu2020; verified via ASEE PEER.

- [17] I Vazquez and S Cutchin. An analysis on pixel redundancy structure in equirectangular images. 2021.
- [18] Hannah Spero, Kenny Miller, Iker Vazquez Lopez, Rezvan Joshaghani, Steven Cutchin, Donna Calhoun, and Josh Enterkine. Communicating dam failure hazards to society through a virtual reality environment of the teton dam failure. In *2021 Undergraduate Research Showcase*, 2021.
- [19] Dylan Eck, Gaby Dagher, Adam Torek, and Steven Cutchin. Untangling many-to-many transactions to aid in taint analysis of the bitcoin blockchain. In *Idaho Conference on Undergraduate Research 2021*, 2021.
- [20] Rebecca M. Broyer, Kenny Miller, Shalini Ramachandran, Sheree Fu, Karen Howell, and Steven Cutchin. Using virtual reality to demonstrate glove hygiene in introductory chemistry labs. *Journal of Chemical Education*, 2020.
- [21] Shalini Ramachandran, Karen Howell, Sheree Fu, and Steve Cutchin. Algorithm bias: A survey of computer science student perceptions. In *Proceedings of the 2020 ASEE PSW Section Conference*, 2020.
- [22] Mike Ramshaw and Steve Cutchin. Reducing motion sickness resulting from movement inside virtual reality environments. In *ICAT-EGVE 2020 — International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments*, pages 141–148, 2020. Verified via DBLP.
- [23] Kurt J. Nilsson, Hilary G. Flint, Yong Gao, Leslie Kendrick, Steve Cutchin, Ryoko Pentecost, and Kristi Pardue. Repetitive head impacts in youth football: Description and relationship to white matter structure. *Sports Health*, 11(6):507–513, 2019. PMID: 31433735.
- [24] Iker Vazquez and Steve Cutchin. Compression Of 16K Video For Mobile VR Playback Over 4K Streams. In Gerd Bruder, Shunsuke Yoshimoto, and Sue Cobb, editors, *ICAT-EGVE 2018 - International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments*. The Eurographics Association, 2018.
- [25] Steve Cutchin and Iker Vazquez. Synchronized shared scene viewing in mixed vr devices in support of group collaboration. In *International Conference on Cooperative Design, Visualization and Engineering*, pages 348–352. Springer, 2016.
- [26] Steve Cutchin and Yuan Li. View dependent tone mapping of hdr panoramas for head mounted displays. In Dirk Reiners, Daisuke Iwai, and Frank Steinicke, editors, *ICAT-EGVE 2016 - International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments*. The Eurographics Association, 2016.
- [27] Iker Vazquez and Steve Cutchin. Synchronized scene views in mixed virtual reality for guided viewing. In *Proceedings of the 26th International Conference on Artificial Reality and Telexistence and the 21st Eurographics Symposium on Virtual Environments*, pages 77–84, 2016.
- [28] Steve Cutchin. Immersion idaho: An undergraduate research program for immersive virtual reality development. In *E-iED 2015: European Immersive Education Summit, Paris France*. Springer Verlag, 2015.
- [29] Neil G Smith, Steve Cutchin, Robert Kooima, Richard A Ainsworth, Daniel J Sandin, Jurgen Schulze, Andrew Prudhomme, Falko Kuester, Thomas E Levy, and Thomas A DeFanti. Cultural heritage omni-stereo panoramas for immersive cultural analyticsfrom the Nile to the Hijaz. In

- Image and Signal Processing and Analysis (ISPA), 2013 8th International Symposium on*, pages 552–557. IEEE, 2013.
- [30] Steve Cutchin. Stereo-panoramic data, March 7 2013. US Patent 20,130,057,643.
 - [31] Steve Smith, Steve Cutchin, Alyn Rockwood, Adel Saad, Neil G Smith, and Thomas E Levy. Demo paper: Virtual and immersive experience presentation of cultural heritage sites. In *Virtual Systems and Multimedia (VSMM), 2012 18th International Conference on*, pages 645–648. IEEE, 2012.
 - [32] April Renee Bailey, Daniel Acevedo, Stephan Vankov, and Steve Cutchin. Kaust campus and visualization laboratory, 2012.
 - [33] ThomasA. DeFanti, Daniel Acevedo, RichardA. Ainsworth, MaxineD. Brown, Steven Cutchin, Gregory Dawe, Kai-Uwe Doerr, Andrew Johnson, Chris Knox, Robert Kooima, Falko Kuester, Jason Leigh, Lance Long, Peter Otto, Vid Petrovic, Kevin Ponto, Andrew Prudhomme, Ramesh Rao, Luc Renambot, DanielJ. Sandin, JurgenP. Schulze, Larry Smarr, Madhu Srinivasan, Philip Weber, and Gregory Wickham. The future of the cave. *Central European Journal of Engineering*, 1(1):16–37, 2011.
 - [34] Jens Schneider, Dina Garatly, Madhusudhanan Srinivasan, Stephen J Guy, Sean Curtis, Steven Cutchin, Dinesh Manocha, Ming C Lin, and Alyn Rockwood. Towards a digital makkah—using immersive 3d environments to train and prepare pilgrims. In *International Conference on Digital Media and its Applications in Cultural Heritage (DMACH)*, pages 1–16, 2011.
 - [35] Todd Margolis, Thomas A DeFanti, Greg Dawe, Andrew Prudhomme, Jurgen P Schulze, and Steve Cutchin. Low cost heads-up virtual reality (huvr) with optical tracking and haptic feedback. In *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, volume 7864, page 41, 2011.
 - [36] Amit Chourasia, Steve Cutchin, and Brad Aagaard. Visualizing the ground motions of the 1906 san francisco earthquake. *Computers & Geosciences*, 34:1798–1805, 2008.
 - [37] A Chourasia, S Cutchin, A Decastro, and G Ely. Visualizing earthquake simulation data. volume 1, page 05, 2007.
 - [38] Amit Chourasia, Steve Cutchin, Yifeng Cui, Reagan W Moore, Kim Olsen, Steven M Day, J Bernard Minster, Philip Maechling, and Thomas H Jordan. Visual insights into high-resolution earthquake simulations. *IEEE Computer Graphics and Applications*, 27(5):28–34, 2007.
 - [39] Amit Chourasia and Jürgen P Schulze. Data centric transfer functions for high dynamic range volume data. Václav Skala-UNION Agency, 2007.
 - [40] Amit Chourasia and Steve Cutchin. $\int \text{em} \int$ digital re-creation of a seven story building shake during an earthquake $\int \text{em} \int$, iee vis scientific animation theatre, sacramento, ca, 2007.
 - [41] Steve Cutchin. Deterministic method and system for determining winners of scratch and win ticket contests and other numeric prize contests, March 6 2007. US Patent App. 11/714,657.
 - [42] A Chourasia and SM Cutchin. Visualization of large and small scale seismic data. volume 1, page 1307, 2006.
 - [43] Steven M Cutchin. La’s future quake, national geographic channel (featuring terashake and puente hills visualizations), 2006.
 - [44] Steven M Cutchin. Recreation of 1906 san francisco earthquake, 100th anniversary earthquake conference, san francisco, 2006.

- [45] Steve Cutchin, Amit Chourasia, Alex Betts, Donna Cox, Stuart Levy, and Robert Patterson. *emj evolution of the universe/emj visualization*, siggraph domefest, los angeles, 2005.
- [46] JR Feramisco, S McMullen, J Sherman, K Hammond, MH Gao, A Decastro, C Chrisman, D Nadeau, J Genetti, S Cutchin, et al. 3-dimensional imaging of muscle and other cell types. 52:S7–S7, 2004.
- [47] J Minster, KB Olsen, R Moore, S Day, P Maechling, T Jordan, M Faerman, Y Cui, G Ely, Y Hu, et al. The sceec terashake earthquake simulation. volume 1, page 05, 2004.
- [48] Steven M Cutchin. *Animations for emj 100 greatest discoveries in chemistry/emj*, discovery science channel, 2004.
- [49] Steven Matthew Cutchin. *Flexible user interface coupling with operation transformation*. PhD thesis, Purdue University, 2000.
- [50] C Bajaj, C Baldazzi, S Cutchin, Alberto Paoluzzi, Valerio Pascucci, and Michele Vicentino. A programming approach for complex animations. part i. methodology. *Computer-Aided Design*, 31(11):695–710, 1999.
- [51] C Bajaj, Steven Cutchin, C Morgia, Alberto Paoluzzi, and Valerio Pascucci. Web based collaborative caad. In *ACM Symposium on Solid and Physical Modeling: Proceedings of the fifth ACM symposium on Solid modeling and applications*, volume 8, pages 326–327, 1999.
- [52] Chandrajit Bajaj and Steven Cutchin. Web based collaborative visualization of distributed and parallel simulation. In *Proceedings of the 1999 IEEE symposium on Parallel visualization and graphics*, pages 47–54, 1999.
- [53] Chandrajit Bajaj, Steven Cutchin, Valerio Pascucci, and Guozhong Zhuang. Error resilient streaming of compressed vrml. *Rapport technique, TICAM, Univ. of Texas*, page 66, 1998.
- [54] C Bajaj, C Baldazzi, S Cutchin, A Paoluzzi, V Pascucci, and M Vicentino. Web-based approach for very complex animations through geometric programming. 1998.
- [55] Chandrajit Bajaj, Steve Cutchin, et al. Web based collaboration-aware synthetic environments. pages 143–150. Citeseer, 1997.
- [56] Chandrajit L Bajaj, Fausto Bernardini, Steven M Cutchin, Kwunnnan Lin, and Elisha Sacks. Comprehensive analysis of joints from patient clinical data. Technical report, 1997.
- [57] Chandrajit L Bajaj, Fausto Berardini, Steven Cutchin, and Raimund Merket. A cscw framework for networked multimodal synthetic environments. Technical report, 1996.
- [58] Chandrajit Bajaj and Steven Cutchin. Collaborative multimedia in shastra. volume 5, pages 365–366, 1995.
- [59] Vinod Anupam, Chandrajit Bajaj, Fausto Bernardini, Steve Cutchin, Jindon Chen, Daniel Schikore, Guoliang Xu, Peinan Zhang, and Weiping Zhang. Scientific problem solving in a distributed and collaborative multimedia environment. *Mathematics and computers in simulation*, 36(4):433–442, 1994.
- [60] Chandrajit Bajaj, Fausto Bernardini, Steve Cutchin, and Kokichi Sugihara. Path-planning for a mobile robot sweeper. Technical report, 1994.
- [61] Vinod Amipam, Chandrajit Bajaj, Steve Cutchin, Susan Evans, Insung Ikm, Jindon Chen, Andrew Royappa, Daniel Schikore, and Guoliang Xu. Scientific problem solving in a distributed and collaborative geometric environment1. page 72, 1993.

- [62] Chandrajit L Bajaj and Steve Cutchin. The gati client/server animation toolkit. In *Communicating with virtual worlds*, pages 413–423. Springer Japan, 1993.
- [63] C Bajaj and S Cutchin. Interactive animation using gati. volume 93, 1992.
- [64] Vinod Anupam, Chanderjit Bajaj, and Steven Klinkner. *The SHILP Solid Modeling and Display Toolkit A User's Manual*, 1990.
- [65] V Anupam, C Bajaj, S Cutchin, T Dey, I Ihm, and S Klinkner. Shilp: Graphical creation, editing and display of algebraic surface models. 1989.
- [66] Suchit Jhunjhunwala, Menno C. van Zelm, Mandy M. Peak, Steve Cutchin, Roy Riblet, Jacques J. M. van Dongen, Frank G. Grosveld, Tobias A. Knoch, and Cornelis Murre. The 3d structure of the immunoglobulin heavy-chain locus: Implications for long-range genomic interactions. *Cell*, 133(2):265–279, 2008/03/24 2008.
- [67] Noriyoshi Osumi, Mikio Shinya, Takeaki Mori, Takeshi Sunaga, C Bajaj, Steve Cutchin, and Raimund Merkert. Nls: Collaborative virtual environment to promote shared awareness. volume 96, pages 41–45.
- [68] C Bajaj and S Cutchin. Web based thin-client architecture for collaborative visualization. Technical report.